

## Lab 12

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Section: A

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### Task 0

```
@23-22412-038-sys → /workspaces/terraform_machine (main) $
```

```
@23-22412-038-sys → /workspaces/terraform_machine (main) $ aws --version  
aws-cli/2.33.2 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
```

```
Setting up terraform (1.4.7) ...  
● @23-22412-038-sys → /workspaces/terraform_machine (main) $ terraform --version  
Terraform v1.4.3  
on linux_amd64  
○ @23-22412-038-sys → /workspaces/terraform_machine (main) $
```

```
@23-22412-038-sys → /workspaces/terraform_machine (main) $ ansible --version  
ansible collection location = /home/codespace/.ansible/collections:/usr/share/a  
nsible/collections  
executable location = /usr/bin/ansible  
python version = 3.12.3 (main, Nov 6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/py  
thon3)  
jinja version = 3.1.6  
libyaml = True  
@23-22412-038-sys → /workspaces/terraform_machine (main) $
```

```
@23-22412-038-sys → /workspaces/terraform_machine (main) $ aws sts get-caller-ide  
ntity  
{  
  "UserId": "623705167686",  
  "Account": "623705167686",  
  "Arn": "arn:aws:iam::623705167686:root"  
}
```

### Task 1

```
Cloning into 'terraform_machine'...  
remote: Enumerating objects: 30, done.  
remote: Counting objects: 100% (30/30), done.  
remote: Compressing objects: 100% (26/26), done.  
remote: Total 30 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (30/30), 7.03 KiB | 7.03 MiB/s, done.  
Resolving deltas: 100% (5/5), done.
```

```

Generating public/private ed25519 key pair.
Created directory '/home/codespace/.ssh'.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Uh7T5J+ajLuZeUKDMZB35qBQ2Y8L33Si+DJHgXih8bw codespace@codespaces-594742
The key's randomart image is:
+--[ED25519 256]--+
|  ..+  .          |
| o = + o+         |
| B * B+ o         |
| o B =o=o...      |
| . =.XSo o        |
| E *.+o o         |
| o ...+           |
+---+
total 20
drwx----- 2 codespace codespace 4096 Jan  6 07:42 .
drwxr-x--- 1 codespace codespace 4096 Jan  6 07:42 ..
-rw----- 1 codespace codespace  419 Jan  6 07:42 id_ed25519
-rw-r--r-- 1 codespace codespace  109 Jan  6 07:42 id_ed25519.pub

```

```

touch terraform.tfvars
ls -la terraform.tfvars
-rw-rw-rw- 1 codespace codespace 0 Jan  6 07:44 terraform.tfvars

```

```

GNU nano 7.2 terraform.tfvars *
vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix           = "dev"
instance_type        = "t3.micro"
public_key           = "~/.ssh/id_ed25519.pub"
private_key          = "~/.ssh/id_ed25519"

```

**Terraform has been successfully initialized!**

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

**Apply complete! Resources: 10 added, 0 changed, 0 destroyed.**

**Outputs:**

```

webserver_public_ips = [
  "158.252.83.133",
  "3.28.136.221",

```

1

## Task 2

```
@23-22412-038-sys →/workspaces/terraform_machine (main) $ terraform plan
terraform apply
```

```
aws_instance.example: Creating...
aws_instance.example: Still creating... [00m10s elapsed]
aws_instance.example: Creation complete after 13s [id=i-0fd45e8764323df23]
```

**Apply complete! Resources: 1 added, 0 changed, 0 destroyed.**

```
ansible --version
'ansible-core' already seems to be installed. Not modifying existing installation in '/usr/local/py-utils/venvs/ansible-core'. Pass
'--force' to force installation.
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/site-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/py-utils/bin/ansible
  python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
```

```
webserver_public_ips = [
    "158.252.83.133",
    "3.28.136.221",
]
```

```
ls -la hosts
-rw-rw-rw- 1 codespace codespace 0 Jan  6 08:01 hosts
```

[Preview] README.md

README.md

hosts

×

workspaces > terraform\_machine > hosts

```
1 158.252.83.133 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
2 3.28.136.221 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

```
[WARNING]: Host '158.252.83.133' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
```

```
158.252.83.133 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

```
[WARNING]: Host '3.28.136.221' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
```

```
3.28.136.221 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

## Task 3

[Preview] README.md

main.tf M X

main.tf

```
13 module "myapp-subnet" {
18   env_prefix = var.env_prefix
19   default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
20 }
21
22 module "myapp-webserver" {
23   source = "./modules/webserver"
24   env_prefix = var.env_prefix
25   instance_type = var.instance_type
26   availability_zone = var.availability_zone
27   public_key = var.public_key
28   my_ip = local.my_ip
29   vpc_id = aws_vpc.myapp_vpc.id
30   subnet_id = module.myapp-subnet.subnet.id
31
32   # Loop count
33   count = 3
34   # Use count.index to differentiate instances
35   instance_suffix = count.index
36 }
```

```
module.myapp-webserver[2].aws_key_pair.ssh-key: Creating...
module.myapp-webserver[2].aws_security_group.web_sg: Creating...
module.myapp-webserver[2].aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverkey-2]
module.myapp-webserver[2].aws_security_group.web_sg: Creation complete after 3s [id=sg-08a64a1ae3f5df444]
module.myapp-webserver[2].aws_instance.myapp-server: Creating...
module.myapp-webserver[2].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Creation complete after 12s [id=i-01e1fcd5d2b3157e2]
```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

```
webserver_public_ips = [
  "158.252.83.133",
  "3.28.136.221",
  "158.252.32.107",
]
```



```
[Preview] README.md  main.tf M  hosts U X
hosts
1  [ec2]
2  158.252.83.133
3  3.28.136.221
4
5  [ec2:vars]
6  ansible_user=ec2-user
7  ansible_ssh_private_key_file=~/.ssh/id_ed25519
8  ansible_ssh_common_args='-o StrictHostKeyChecking=no'
9
10 [droplet]
11 158.252.32.107
12
13 [droplet:vars]
14 ansible_user=ec2-user
15 ansible_ssh_private_key_file=~/.ssh/id_ed25519
16 ansible_ssh_common_args='-o StrictHostKeyChecking=no'
```

```
[WARNING]: Host '158.252.83.133' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.83.133 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.136.221' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.136.221 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

```
[WARNING]: Host '158.252.83.133' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.83.133 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

## Task 4

```
1  [defaults]
2  host_key_checking = False
3  interpreter_python = /usr/bin/python3
4  |
```

[Preview] README.md

main.tf M

hosts U X

.ansible.cfg

≡ hosts

1

[ec2]

2

158.252.83.133

3

3.28.136.221

4

5

[ec2:vars]

6

ansible\_user=ec2-user

7

ansible\_ssh\_private\_key\_file=~/.ssh/id\_ed25519

8

9

[droplet]

10

158.252.32.107

11

|

12

[droplet:vars]

13

ansible\_user=ec2-user

14

ansible\_ssh\_private\_key\_file=~/.ssh/id\_ed25519

158.252.32.107 | SUCCESS => {

"changed": false,

"ping": "pong"

}

158.252.83.133 | SUCCESS => {

"changed": false,

"ping": "pong"

}

3.28.136.221 | SUCCESS => {

"changed": false,

"ping": "pong"

}

[Preview] README.md

main.tf M

hosts U

! my-playbook.yaml U X

.ansible.cfg

! my-playbook.yaml

1

---

2

- name: Configure nginx web server

3

hosts: ec2

4

become: true

5

tasks:

6

- name: install nginx and update cache

7

yum:

8

name: nginx

9

state: present

10

update\_cache: yes

11

12

- name: start nginx server

13

service:

14

name: nginx

15

state: started

```
PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [158.252.83.133]
ok: [3.28.136.221]

TASK [install nginx and update cache] *****
changed: [158.252.83.133]
changed: [3.28.136.221]

TASK [start nginx server] *****
changed: [158.252.83.133]
changed: [3.28.136.221]

PLAY RECAP *****
158.252.83.133      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
3.28.136.221      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

## Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](http://nginx.org).  
Commercial support is available at [nginx.com](http://nginx.com).

*Thank you for using nginx.*

[Preview] README.md

main.tf M

hosts U

! my-playbook.yaml U X

ansible.cfg

! my-playbook.yaml

```
1 ---
2 - name: Configure nginx web server
3   hosts: droplet
4   become: true
5   tasks:
6     - name: install nginx and update cache
7       yum:
8         name: nginx
9         state: present
10        update_cache: yes
11
12    - name: start nginx server
13      service:
14        name: nginx
15        state: started
```

## Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](http://nginx.org).  
Commercial support is available at [nginx.com](http://nginx.com).

*Thank you for using nginx.*

```
ls -la ansible.cfg
-rw-rw-rw- 1 codespace codespace 0 Jan  6 08:52 ansible.cfg
```

[Preview] README.md

main.tf M

hosts U

! my-playbook.yaml U

ansible.cfg X

```
home > codespace > ansible.cfg
1  [defaults]
2  host_key_checking=False
3  interpreter_python = /usr/bin/python3
```

[Preview] README.md

main.tf M X

hosts U

! my-playbook.yaml U

ansible.cfg

```
main.tf
13 module "myapp-subnet" {
20 }
21
22 module "myapp-webserver" {
23     source = "../modules/webserver"
24     env_prefix = var.env_prefix
25     instance_type = var.instance_type
26     availability_zone = var.availability_zone
27     public_key = var.public_key
28     my_ip = local.my_ip
29     vpc_id = aws_vpc.myapp_vpc.id
30     subnet_id = module.myapp-subnet.subnet.id
31
32     # Loop count
33     count = 1
34     # Use count.index to differentiate instances
35     instance_suffix = count.index
36 }
```

```
webserver_public_ips = [
    "158.252.83.133",
]
```

[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform\_machine), ignoring it as an ansible.cfg  
[https://docs.ansible.com/ansible/devel/reference\\_appendices/config.html#cfg-in-world-writable-dir](https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir)

```
PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [158.252.83.133]

TASK [install nginx and update cache] *****
ok: [158.252.83.133]

TASK [install openssl] *****
ok: [158.252.83.133]

TASK [start nginx server] *****
changed: [158.252.83.133]

PLAY RECAP *****
158.252.83.133      : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



# Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to [nginx.org](https://nginx.org).  
Commercial support is available at [nginx.com](https://nginx.com).

*Thank you for using nginx.*

```
ok: [158.252.83.133]

TASK [Create SSL directories] *****
changed: [158.252.83.133] => (item={'path': '/etc/ssl/private', 'mode': '0700'})
changed: [158.252.83.133] => (item={'path': '/etc/ssl/certs', 'mode': '0755'})

TASK [Get IMDSv2 token] *****
ok: [158.252.83.133]

TASK [Get current public IP] *****
ok: [158.252.83.133]

TASK [Show current public IP] *****
ok: [158.252.83.133] => {
  "msg": "Public IP is 158.252.83.133"
}

TASK [Generate self-signed SSL certificate] *****
changed: [158.252.83.133]

PLAY RECAP *****
158.252.83.133      : ok=8    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



```
TASK [Gathering Facts] *****
ok: [158.252.83.133]

TASK [install php-fpm and php-curl] *****
changed: [158.252.83.133]

TASK [Copy website files] *****
changed: [158.252.83.133]

TASK [Copy nginx.conf template] *****
changed: [158.252.83.133]

TASK [Restart nginx] *****
changed: [158.252.83.133]

TASK [Start and enable php-fpm] *****
changed: [158.252.83.133]

PLAY RECAP *****
158.252.83.133      : ok=14   changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Not secure <https://158.252.83.133>



## Your connection is not private

Attackers might be trying to steal your information from **158.252.83.133** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET::ERR\_CERT\_AUTHORITY\_INVALID



[Turn on enhanced protection](#) to get Chrome's highest level of security

Advanced

Back to safety

## Nginx Front End Web Server

|              |                                                        |
|--------------|--------------------------------------------------------|
| Hostnames:   | ip-10-0-10-107.mt-central-1.compute.internal           |
| Instance ID: | i-0cafb7153b7a7b7ab                                    |
| Private IP:  | 10.0.10.107                                            |
| Public IP:   | 158.252.83.133                                         |
| Public DNS:  | https://N/A                                            |
| Deployed:    | 2026-01-06 12:04:23                                    |
| Status:      | <input checked="" type="checkbox"/> Active and Running |
| Managed By:  | Terraform + Ansible                                    |

## Task 8

```
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 0s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0c3680394866245c8]
aws_vpc.myapp_vpc: Destruction complete after 0s

Destroy complete! Resources: 7 destroyed.
```

```
module.myapp-webserver[0].aws_instance.myapp_server: Creation complete after 12s [id=i-00007c37a1b1b270]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "3.29.5.229",
]
```

```

  ____  _
 / ___|| | | |
 \___ \| |_| |
  ___) |  __/
 |____|_| |_|

#_
##### Amazon Linux 2023
#####\
\###|
\#/
Vv' -.-> https://aws.amazon.com/linux/amazon-linux-2023
  _
 _/m/

```

## Task 9



```

compose.yaml
1  - name: Deploy Docker Containers
2    hosts: all
3    become: true
4    user: "{{ normal_user }}"
5    vars_files:
6      - project-vars.yaml
7    tasks:
8      - name: check if docker-compose file exists
9        stat:
10         path: /home/{{ normal_user }}/compose.yaml
11         register: compose_file
12
13      - name: copy docker-compose file
14        copy:
15         src: "{{ docker_compose_file_location }}/compose.yaml"
16         dest: /home/{{ normal_user }}/compose.yaml
17         mode: '0644'
18         when: not compose_file.stat.exists
19
20      - name: deploy containers using docker-compose
21        command: docker compose up -d
22        register: compose_result
23        changed_when: "'Creating' in compose_result.stdout or 'Recreating' in compose_result.stdout"

```

```

TASK [verify docker access] *****
ok: [3.29.5.229]

TASK [display docker ps output] *****
ok: [3.29.5.229] => {
  "docker_ps.stdout": "CONTAINER ID   IMAGE          COMMAND                  CREATED   STATUS    PORTS    NAMES"
}

PLAY [Deploy Docker Containers] *****

TASK [Gathering Facts] *****
ok: [3.29.5.229]

TASK [check if docker-compose file exists] *****
ok: [3.29.5.229]

TASK [copy docker-compose file] *****
changed: [3.29.5.229]

TASK [deploy containers using docker-compose] *****
ok: [3.29.5.229]

PLAY RECAP *****
3.29.5.229          : ok=14   changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

```

      + cidr_blocks      = [
        + "0.0.0.0/0",
      ]
      + from_port        = 80
      + ipv6_cidr_blocks = []
      + prefix_list_ids  = []
      + protocol         = "tcp"
      + security_groups  = []
      + self             = false
      + to_port          = 80
    },
  ],
  name           = "dev-web-sg-0"
  tags           = {
    "Name" = "dev-default-sg"
  }
  # (9 unchanged attributes hidden)
}

```

Plan: 0 to add, 1 to change, 0 to destroy.

module.myapp-webserver[0].aws\_security\_group.web\_sg: Modifying... [id=sg-01c0dec942001d806]

module.myapp-webserver[0].aws\_security\_group.web\_sg: Modifications complete after 0s [id=sg-01c0dec942001d806]

**Apply complete! Resources: 0 added, 1 changed, 0 destroyed.**

**Outputs:**

```

webserver_public_ips = [
  "3.29.5.229",
]

```

3.29.5.229:3000

Gitea HTTP Listen Port \*

3000

Port number the Gitea web server will listen on.

Gitea Base URL \*

http://3.29.5.229:3000/

Base address for HTTP(S) clone URLs and email notifications.

Log Path \*

/data/gitea/log

Log files will be written to this directory.

☐ Enable Update Checker

Checks for new version releases periodically by connecting to gitea.io.

### Optional Settings

► Email Settings

► Server and Third-Party Service Settings

► Administrator Account Settings

These configuration options will be written into: `/data/gitea/conf/app.ini` 

Install Gitea

## Task 10:

```
main.tf
37 resource "null_resource" "configure_server" {}
38   triggers = {
39     # Is se Terraform ko pata chalega ke naya IP kab aaya
40     webserver_public_ips_for_aws = join(",", module.myapp-webserver.webserver_public_ips)
41   }
42
43   depends_on = [module.myapp-webserver]
44
45   provisioner "local-exec" {
46     command = <<-EOT
47     ansible-playbook -i ${self.triggers.webserver_public_ips_for_aws}, \
48       --private-key "${var.private_key_location}" --user ec2-user \
49       my-playbook.yaml
50     EOT
51   }
52
53
```

Destroy complete! Resources: 0 destroyed.

```
null_resource.configure_server (local-exec): skipping: no hosts matched
null_resource.configure_server (local-exec): PLAY [Adding user to docker group] *****
null_resource.configure_server (local-exec): skipping: no hosts matched
null_resource.configure_server (local-exec): PLAY [Deploy Docker Containers] *****
null_resource.configure_server (local-exec): skipping: no hosts matched
null_resource.configure_server (local-exec): PLAY RECAP *****
null_resource.configure_server: Creation complete after 1s [id=1695432964614148986]
```

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

#### Outputs:

```
webservers_public_ips = [
  "40.172.121.214",
]
```

40.172.121.214:3000

Gitea Base URL \*

http://40.172.121.214:3000/

Base address for HTTP(S) clone URLs and email notifications.

Log Path \*

/data/gitea/log

Log files will be written to this directory.

☐ Enable Update Checker

Checks for new version releases periodically by connecting to gitea.io.

Optional Settings

► Email Settings

► Server and Third-Party Service Settings

► Administrator Account Settings

These configuration options will be written into:

/data/gitea/conf/app.ini

Install Gitea

## task 11

```
ls -la inventory_aws_ec2.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan  6 13:58 inventory_aws_ec2.yaml
```



Apply complete! Resources: 3 added, 1 changed, 1 destroyed.

#### Outputs:

```
prod-webserver_public_ips = [
    "3.28.45.153",
]
webserver_public_ips = [
    "40.172.121.214",
]
```

```
prod-webserver_public_ips = [
    "3.28.45.153",
]
webserver_public_ips = [
    "40.172.121.214",
]
```

```
Collecting boto3
  Downloading boto3-1.42.22-py3-none-any.whl.metadata (6.8 kB)
Collecting botocore
  Downloading botocore-1.42.22-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3)
  Downloading jmespath-1.0.1-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/codespace/.local/lib/python3.12/site-packages (from botocore
ost0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/codespace/.local/lib/python3.12/site-packages (from botocore)
Requirement already satisfied: six>=1.5 in /home/codespace/.local/lib/python3.12/site-packages (from python-dateutil<3.0.0,>=2.1
) (1.17.0)
Downloading boto3-1.42.22-py3-none-any.whl (140 kB)
Downloading botocore-1.42.22-py3-none-any.whl (14.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 14.6/14.6 MB 27.8 MB/s 0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.42.22 botocore-1.42.22 jmespath-1.0.1 s3transfer-0.16.0
```

```
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils.text' is deprecated. This feature will be removed from ansible-
version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils.text' is deprecated. This feature will be removed from ansibl
re version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing 'disable_lookups' to 'template' is deprecated. This feature will be removed from ansible-core version 2
@all:
|--@ungrouped:
|--@aws_ec2:
| |--ip-10-0-1-55.me-central-1.compute.internal
| |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
| |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
| |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
| |--ip-10-0-10-197.me-central-1.compute.internal
| |--ip-10-0-10-21.me-central-1.compute.internal
| |--ip-10-0-1-61.me-central-1.compute.internal
```

## Task 12

version 2.24. Use ansible.module\_utils.common.text.converters instead.  
 [DEPRECATION WARNING]: Importing 'to\_native' from 'ansible.module\_utils.\_text' is deprecated. This feature will be removed in version 2.24. Use ansible.module\_utils.common.text.converters instead.  
 [DEPRECATION WARNING]: Passing 'disable\_lookups' to 'template' is deprecated. This feature will be removed from ansible

```
@all:
|  |--@ungrouped:
|  |--@aws_ec2:
|  |  |--ip-10-0-1-55.me-central-1.compute.internal
|  |  |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
|  |  |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
|  |  |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
|  |  |--ip-10-0-10-197.me-central-1.compute.internal
|  |  |--ip-10-0-10-21.me-central-1.compute.internal
|  |  |--ip-10-0-1-61.me-central-1.compute.internal
|  |--@tag_Name_dev_ec2_instance_2:
|  |  |--ip-10-0-1-55.me-central-1.compute.internal
|  |--@tag_Name_prod_ec2_instance_0:
|  |  |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
|  |--@tag_Name_MyServer:
|  |  |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
|  |--@tag_Name_dev_ec2_instance_0:
|  |  |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
|  |  |--ip-10-0-10-21.me-central-1.compute.internal
|  |--@tag_Name_dev_ec2_instance_1:
|  |  |--ip-10-0-10-197.me-central-1.compute.internal
|  |  |--ip-10-0-1-61.me-central-1.compute.internal
```

locals.tf M index.php U nginx.confj2 U .ansible.cfg ! inventory\_aws\_ec2.yaml U ×

```
! inventory_aws_ec2.yaml
1  ---
2  plugin: aws_ec2
3  regions:
4  |  - me-central-1
5  keyed_groups:
6  |  - key: tags
7  |    prefix: tag
8  |    separator: "_"
9  |  - key: instance_type
10 |    prefix: instance_type
11 |    separator: "_"
```

```
|  |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
|  |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
|  |--ip-10-0-10-197.me-central-1.compute.internal
|  |--ip-10-0-10-21.me-central-1.compute.internal
|  |--ip-10-0-1-61.me-central-1.compute.internal
|--@tag_Name_prod_ec2_instance_0:
|  |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
|--@instance_type_t3_nano:
|  |--ec2-3-28-45-153.me-central-1.compute.amazonaws.com
|--@tag_Name_MyServer:
|  |--ec2-3-28-135-61.me-central-1.compute.amazonaws.com
|--@tag_Name_dev_ec2_instance_0:
|  |--ec2-40-172-121-214.me-central-1.compute.amazonaws.com
|  |--ip-10-0-10-21.me-central-1.compute.internal
|--@tag_Name_dev_ec2_instance_1:
|  |--ip-10-0-10-197.me-central-1.compute.internal
|  |--ip-10-0-1-61.me-central-1.compute.internal
```

```

TASK [Save public IP as fact for template] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Generate self-signed SSL certificate] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

PLAY [Deploy Nginx website and configuration files] *****

TASK [Gathering Facts] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [install php-fpm and php-curl] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Copy website files] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Copy nginx.conf template] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Restart nginx] *****
changed: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

TASK [Start and enable php-fpm] *****
ok: [ec2-3-28-45-153.me-central-1.compute.amazonaws.com]

PLAY RECAP *****
ec2-3-28-45-153.me-central-1.compute.amazonaws.com : ok=17  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

```

## Task 13

```

ansible-galaxy role init nginx
ansible-galaxy role init ssl
ansible-galaxy role init webapp
cd ..
ls -R
- Role nginx was created successfully
- Role ssl was created successfully
- Role webapp was created successfully
.:

```



```
..
ansible.cfg inventory my-playbook.yaml roles

./inventory:

./roles:
nginx ssl webapp

./roles/nginx:
README.md defaults files handlers meta tasks templates tests vars

./roles/nginx/defaults:
main.yml

./roles/nginx/files:

./roles/nginx/handlers:
main.yml

./roles/nginx/meta:
```

[WARNING]: Ansible is being run in a world-writable directory (/workspaces/terraform-machine/ansible), ignoring it as an ansible.cfg source. For more information see [https://docs.ansible.com/ansible/devel/reference\\_appendices/config.html#cfg-in-world-writable-dir](https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir)

```
PLAY [Deploy NGINX Web Stack with SSL and PHP] *****

TASK [Gathering Facts] *****
ok: [40.172.121.214]

TASK [nginx : Install nginx] *****
ok: [40.172.121.214]

TASK [nginx : Install openssl] *****
ok: [40.172.121.214]

TASK [nginx : Start and enable nginx] *****
ok: [40.172.121.214]

PLAY RECAP *****
40.172.121.214      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
TASK [ssl : Generate self-signed certificate] *****
ok: [40.172.121.214]

TASK [webapp : Install PHP packages] *****
ok: [40.172.121.214]

TASK [webapp : Copy PHP website] *****
changed: [40.172.121.214]

TASK [webapp : Deploy nginx config] *****
changed: [40.172.121.214]

TASK [webapp : Start and enable php-fpm] *****
ok: [40.172.121.214]

RUNNING HANDLER [webapp : Restart nginx] *****
changed: [40.172.121.214]

PLAY RECAP *****
40.172.121.214      : ok=15   changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



## Your connection is not private

Attackers might be trying to steal your information from **40.172.121.214** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

NET::ERR\_CERT\_AUTHORITY\_INVALID



[Turn on enhanced protection](#) to get Chrome's highest level of security

Advanced

Back to safety

 **Nginx Front End Web Server**

|              |                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|
| Hostname:    | ip-10-0-10-133.mtc-central-1.compute.internal                                                          |
| Instance ID: | i-07f94bc01445483c7                                                                                    |
| Private IP:  | 10.0.10.133                                                                                            |
| Public IP:   | 40.172.121.214                                                                                         |
| Public DNS:  | https://N/A                                                                                            |
| Deployed:    | 2020-01-07 07:26:46                                                                                    |
| Status:      |  Active and Running |
| Managed By:  | Terraform + Ansible                                                                                    |

Clean up:



Either you have not created any objects yet or the existing objects were already deleted outside of Terraform.

**Destroy complete! Resources: 0 destroyed.**

```
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 1,
  "lineage": "0a7b60e6-94d2-a791-71ca-195066bed5e5",
  "outputs": {},
  "resources": [],
  "check_results": null
}
```